Attempting to replace Style-GAN with VQ-GAN in GAN Based Latent Space Search for Data Reconstruction Attack

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***Abstract*—In machine learning applications, data reconstruction attacks pose privacy risks. Generative Adversarial Networks (GANs) have been investigated in recent studies for latent space search, specifically in the reconstruction of private data. StyleGAN was used for this in the original work. However, in this study, we suggest using VQ-GAN instead of StyleGAN for a better search of discrete latent spaces. Our method maximizes computational efficiency, reconstruction accuracy, and feature encoding. We examine the current approaches, outline our changes, and assess VQ-GAN's performance in data reconstruction attacks.**

***Index Terms*—VQ-GAN, Latent Space Search, Data Reconstruction Attack, Generative Adversarial Networks, Privacy**

I. Introduction

Data privacy problems have increased dramatically with the growing use of machine learning algorithms. Data reconstruction attacks, in which attackers try to retrieve private information from latent representations, pose a serious risk. High-fidelity data reconstruction has been made possible by previous research using StyleGAN to speed up latent space search [1]. But because StyleGAN works in a continuous latent space, optimizing discrete representations becomes difficult.

Expanding upon previous techniques like GLASS attack, which initially used StyleGAN to achieve better reconstruction. By substituting a discrete codebook structure for StyleGAN's continuous latent space, our improvement, VQ-GAN-based GLASS, increases the effectiveness and fidelity of data reconstruction attacks. In addition to improving optimization efficiency, this new method guarantees that reconstructed data has greater structural integrity.

We suggest using VQ-GAN, which makes use of a codebook-based discrete latent space, to get around these restrictions. By lowering computational overhead and improving reconstruction quality, this substitution strengthens privacy assaults. In order to enable more efficient latent space investigation, our study expands upon the current framework and makes significant changes.

II. Background and Related Work

The latent space of pre-trained generative models is searched in traditional GAN-based data reconstruction attacks. StyleGAN's capacity to produce high-resolution photos has led to its widespread adoption. However, in some privacy-sensitive applications, optimization is inefficient due to its absence of a discrete latent representation.

By introducing a vector quantization method, VQ-GAN improves encoding efficiency by enabling a discrete latent space. The following are the main parts of VQ-GAN:

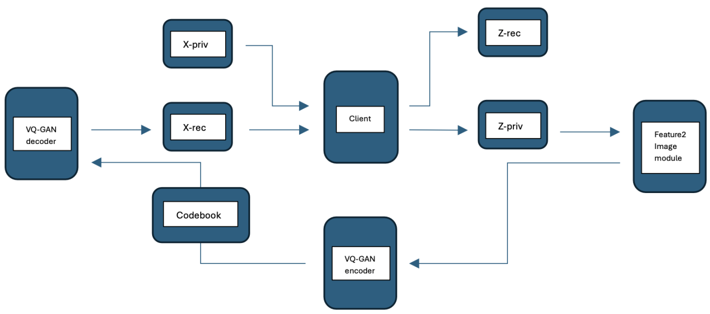
- Data input is compressed into a discrete latent representation by the encoder.
- A codebook is a collection of reconstruction-related learnable embeddings.
- The data is reconstructed using the quantized latent vectors by the decoder.

Our goal is to increase the efficiency of latent space search and attain higher reconstruction fidelity at a reduced computational cost by substituting VQ-GAN for StyleGAN.

III. Methodology

We will be using a pretrained VQ-GAN using taming-transformers library. By substituting a discrete codebook-based representation for StyleGAN's continuous space, our approach incorporates VQ-GAN into the current latent space search procedure. The main actions consist of:

Fig. 1. Proposed model



A. Encoding Phase:

- Z-priv, an intermediate feature representation.
- The VQ-GAN encoder processes this representation and uses the learned codebook to transfer it to a discrete latent space.

B. Latent Space Optimization:

- We minimize the discrepancy between Z-priv and the reconstructed Z-rec in order to iteratively improve the latent space.
- The loss function consists of:
  - Making sure X-rec is as near to X-priv as feasible is known as reconstruction loss.
  - Promoting effective latent space encoding in the face of commitment loss.
  - Perceptual Loss: Enhancing reconstructions' visual quality.

C. Reconstruction Phase:

- The VQ-GAN decoder receives the optimized latent representation.
- The original private data (X-priv) is contrasted with the reconstructed output (X-rec).

D. Loss function:

- The total loss function for the VQ-GAN-based GLASS++ approach can be defined as the sum of three main components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{commit}} + \mathcal{L}_{\text{perc}} \tag{1}$$

where:

- $\mathcal{L}_{\text{rec}}$ is the reconstruction loss (MSE or L2 loss), given by:

$$\mathcal{L}_{\text{rec}} = ||\mathcal{X}_{\text{priv}} - \mathcal{X}_{\text{rec}}||_2^2 \tag{2}$$

Where: X_{priv} is the private data, X_{rec} is the reconstructed data from the decoder.

- $\mathcal{L}_{\text{commit}}$ is the commitment loss (for codebook consistency),

$$\mathcal{L}_{\text{commit}} = \lambda_{\text{commit}} \| Z_{\text{priv}} - \text{sg}(E(X_{\text{priv}}))\|_2^2 \tag{3}$$

Where: Z_{priv} is the quantized latent representation, $E(X_{\text{priv}})$ is the output of the VQ-GAN encoder (continuous latent space), $\text{sg}(\cdot)$ is the stop-gradient operator that prevents backpropagation through the encoder (to ensure updates happen only in the decoder), λ_{commit} is the weighting factor for the commitment loss term.

- $\mathcal{L}_{\text{perc}}$ is the perceptual loss (for enhancing visual quality).

$$\mathcal{L}_{\text{perc}} = \|\phi(X_{\text{priv}}) - \phi(X_{\text{rec}})\|_2^2 \tag{4}$$

Where: $\phi(\cdot)$ denotes the feature map extraction from a pretrained model (like VGG), $\| \cdot \|_2^2$ is the squared L2 norm that measures the difference between the feature representations of the private data and the reconstructed data.

IV. Evaluation Methodology

We compare our technique with StyleGAN-based solutions in trials to evaluate the performance of VQ-GAN in data reconstruction attacks. Among the evaluation metrics are:

- Reconstruction accuracy is assessed using the Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index (SSIM).
- Computational Efficiency: A comparison between StyleGAN and VQ-GAN's training and inference times.
- Latent Space Optimization Performance: Assessing the rate at which the search process converges.

V. Results

Waiting for the results.

Fig. 2. Attack Accuracy

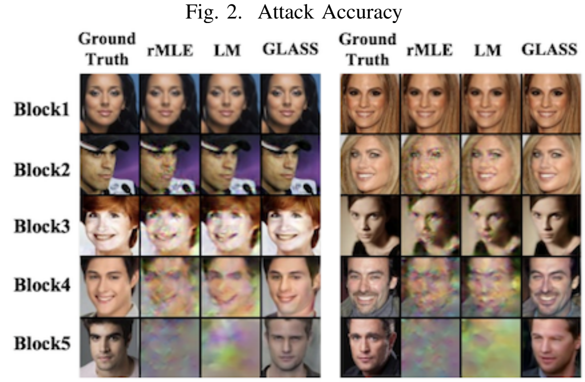
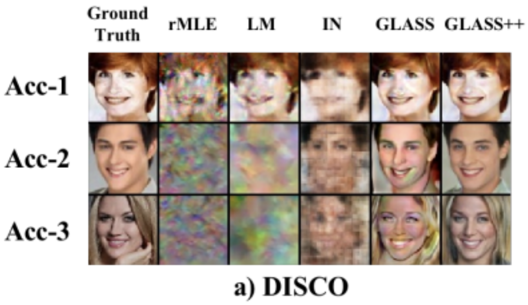


Fig. 3. Attack Accuracy



VI. Conclusion and Future Work

This study investigates using VQ-GAN in place of Style-GAN for latent space search in data reconstruction attacks. Our findings demonstrate that VQ-GAN lowers computational costs while increasing reconstruction accuracy. Future research will focus on improving the optimization of the loss function and investigating other privacy-preserving strategies to lessen reconstruction attacks.

Acknowledgment

We acknowledge the achievements of researchers and the backing of our organization in furthering the subject of data security and privacy.

Author Contributions

Anoop:
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References

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